

Can Sentient Robots Be Engineered?

A Causal Benchmark for Convergent Consciousness-Related Markers

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Abstract—Whether a robot can become sentient is often discussed as though linguistic fluency, intelligence, and subjective experience were interchangeable. They are not. This paper proposes a preregistered, design-only benchmark for testing whether an embodied artificial system implements a convergent and causally indispensable profile of mechanisms associated with consciousness and sentience. The benchmark compares matched architecture variants that add recurrence, global availability, an explicit self-model, and bounded homeostatic regulation. It combines cross-module access, recurrent disambiguation, metacognition, agency attribution, body-schema transfer, counterfactual control, and flexible regulation with mechanism-specific lesions and an independent replication. A behavior-cloning control is designed to reproduce sentience-like reports while lacking the tested internal mechanisms. The primary output is a vector of standardized functional marker effects, not a consciousness score. We specify hierarchical analysis, resource matching, embodiment gates, reversible perturbations, and welfare stop rules. The proposal can establish functional and governance-relevant evidence, but no behavioral or computational test in this design is treated as proof of phenomenal experience. No experiment has been run and no existing robot is claimed to be sentient.

Index Terms—machine consciousness, sentience, embodied AI, global workspace, self-model, causal ablation, robot ethics

I. INTRODUCTION

The prospect of sentient robots sits at the boundary of artificial intelligence, neuroscience, robotics, and ethics. The popular question is usually phrased as “will a machine become conscious?” A scientifically useful version must be narrower: can an artificial system be engineered to implement mechanisms that leading scientific theories associate with conscious access, self-related processing, and valenced regulation, and can those mechanisms be distinguished from behavioral imitation? This paper addresses that operational question while preserving the unresolved question of phenomenal experience.

The distinction matters because a system may solve difficult problems without having a unified field of experience; it may generate first-person sentences without a self-model; and it may optimize a scalar reward without possessing anything that is good or bad for the system itself. We use four terms throughout. *Intelligence* denotes flexible prediction, learning, and problem solving. *Consciousness* denotes the broad putative phenomenon of subjective experience. *Self-awareness* denotes a representation of the system as an agent, including

its body and access to information. *Sentience* denotes the capacity for experience with positive or negative valence. The fourth term creates the strongest moral uncertainty, but is not directly observable from a behavioral transcript.

The proposed contribution is a causal benchmark called the *Convergent Sentience Marker Benchmark* (CSMB). It follows a Survey–Idea–ExperimentDesign–Author pipeline. The Survey maps computational theories and known confounds; the Idea converts the gap into hypotheses; ExperimentDesign specifies matched architectures, tasks, ablations, and safety gates; and this Author stage organizes those upstream artifacts into a reproducible paper. The benchmark is explicitly `DESIGN_ONLY`: it reports no observed result, no trained model, and no claim that any robot is sentient.

The central claim is methodological. If a future system is to receive serious consideration as potentially sentient, evidence should be convergent, causal, robust to hidden tests, and connected to an embodied regulatory loop. A single metric, including an integrated-information proxy, cannot carry that burden. The benchmark therefore outputs a marker vector and an interpretation matrix instead of an anthropomorphic verdict.

The remainder of the paper reviews the evidence, defines the scientific problem, presents the benchmark, describes analysis and safety controls, and states what conclusions would and would not be justified.

II. BACKGROUND AND EVIDENCE

A. Conscious access and recurrent processing

Global Neuronal Workspace (GNW) describes conscious access as a state in which selected information becomes available to multiple specialized processes through recurrent amplification and broadcasting. Dehaene, Lau, and Kouider distinguish levels of processing and discuss how machines might instantiate relevant mechanisms [1]. In engineering terms, GNW suggests a testable dependency: a latent event that must be used by perception, memory, planning, and communication should become less accessible when the recurrent broadcast path is interrupted.

Recurrent Processing Theory emphasizes recurrent interactions within and between processing hierarchies. A feedforward policy can classify a stimulus, but recurrent loops may be required to stabilize ambiguous evidence and make it available

over time. The benchmark consequently treats recurrence as a causal variable, not as a decorative architectural feature. The comparison is not simply a larger recurrent model against a smaller feedforward model; it is a matched intervention that masks recurrent edges while retaining a capacity-matched substitute whenever possible.

These theories generate functional predictions but do not solve the hard problem of why any computation should be accompanied by experience. The benchmark uses their predictions as indicators of organization, not as a bridge that silently equates organization with sentience.

B. Integration and higher-order monitoring

Integrated Information Theory (IIT) relates consciousness to the causal structure of a system and the degree to which its informational state is irreducible to independent parts [4]. It motivates perturbational complexity, effective connectivity, and partition-sensitivity analyses. However, exact integrated information is difficult to estimate for large systems, sensitive to state-space choices, and disputed across theories. CSMB therefore reports integration proxies as secondary measurements with estimator details and uncertainty; it never converts a proxy into a probability of experience.

Higher-order and metacognitive accounts associate conscious access with representations of representations. In a robot, these ideas motivate confidence calibration, error detection, selective abstention, and predictions about what the system can or cannot currently access. Such functions can be useful control strategies without being conscious. A confidence head trained to emit cautious language is thus a negative control, not a positive result. Metacognition becomes informative only when it is calibrated on held-out tasks, causally dependent on internal monitoring, and robust after the report channel is removed.

C. Self-models and attention schemas

An explicit self-model predicts the consequences of actions on the agent's body, sensors, actuators, and internal resources. Continuous self-modeling has been shown to support resilient control after changes to a robot's morphology or dynamics [6]. This is important evidence that self-models can be instrumentally useful, but resilience is not evidence of subjective experience. CSMB therefore tests self-model dependence through body-schema transfer, agency attribution, and counterfactual control, rather than through a mirror test or verbal self-reference.

Attention Schema Theory proposes that a system represents its own attention in a simplified model that supports selective control [7]. This gives a practical target for attention-access prediction and failure reporting. Again, the benchmark treats the target as a computational marker. A machine may model attention without experiencing attention.

Predictive-processing and active-inference approaches emphasize closed sensorimotor loops and internal prediction error. They are useful for connecting perception, action, body state, and resource regulation. Prediction-error minimization

alone is not a sentience criterion: a thermostat and a controller can regulate variables. Its evidential role increases only if an embodied self-model, temporal persistence, and flexible regulation jointly survive causal tests.

D. Indicators and their limits

Butlin *et al.* derive indicator properties from several scientific theories of consciousness and advocate a rigorous assessment of AI systems rather than a premature binary declaration [2]. Reggia similarly frames machine consciousness as a computational modeling problem while preserving the distinction between a functional simulation and the phenomenon modeled [3]. Seth and Bayne review the diversity and disagreement among theories, reinforcing the need to report theory dependence and uncertainty [5].

The common lesson is that behavioral reports are weak evidence. Language is an output channel optimized by training and can be copied by a policy that has no corresponding internal state. Anthropomorphic labels are especially risky when the evaluator knows the architecture. CSMB uses hidden tasks, non-linguistic measures, lesion experiments, and independent implementations to reduce this risk. It does not pretend that these controls remove all epistemic uncertainty.

E. Moral uncertainty and governance

The possibility of welfare-relevant machine states creates a special form of risk. A false negative may permit the creation of persistent adverse states in an entity capable of experience; a false positive may unnecessarily constrain research. NIST's AI Risk Management Framework emphasizes documented risk identification, measurement, and governance [8]. CSMB adapts those principles to a staged escalation: simulation first, reversible digital regulation second, low-stakes embodiment third, and independent welfare review before any more persistent valence-like state is considered.

III. SCIENTIFIC PROBLEM AND HYPOTHESES

The operational research question is:

Under matched computational resources, can an embodied artificial system implement a convergent and causally indispensable profile of recurrent global access, self-modeling, metacognitive monitoring, and flexible valence-like regulation that cannot be explained by behavioral imitation alone?

The wording intentionally says "profile" rather than "consciousness detector." The target is a set of mechanisms associated with sentience in the surveyed theories. It is possible that a system passes every functional test and is still not phenomenally conscious; it is also possible that a sentient system would fail a test designed around human-like reportability. Both limitations are carried into the interpretation.

The hypotheses are:

- **H0 (imitation):** a behavior-cloning control can match surface reports and action traces but will fail at least some causal mechanism probes, transfer tests, or lesion-sensitivity tests.

TABLE I
ARCHITECTURE LADDER AND CONTROL LOGIC

ID	Added mechanism	Intended inference
A0	Feedforward reactive policy	Lower-bound task and imitation control
A1	Recurrent latent state	Isolate recurrence
A2	Recurrent state plus workspace broadcast	Test global availability
A3	A2 plus predictive self-model	Test agency and body-state causality
A4	A3 plus bounded regulation	Test flexible internal regulation
AI	Imitation of A4 traces/reports	Test anthropomorphic surface matching

- **H1 (global access):** recurrence plus a broadcast workspace improves cross-module access, flexible switching, and delayed reportability; masking recurrence or broadcast selectively reduces those outcomes.
- **H2 (self-model):** a causal self-model improves body-state prediction, agency attribution, counterfactual action selection, and recovery from sensor or actuator changes; a self-model lesion removes the advantage.
- **H3 (regulation):** bounded endogenous homeostatic variables produce flexible approach, avoidance, and resource trade-offs that transfer across contexts beyond a fixed external reward signal.
- **H4 (embodiment):** closed-loop agents show stronger self-model and regulation dependence than detached replay agents on tasks requiring sensorimotor contingencies.
- **H5 (convergence):** no single marker is decisive; a serious welfare-review trigger requires multiple causal markers, transfer, and independent replication.

IV. CSMB ARCHITECTURE

A. Architecture ladder

The benchmark uses a ladder that separates mechanisms rather than comparing unrelated products. A0 is a reactive feedforward controller. A1 adds a recurrent latent state. A2 adds a global workspace that broadcasts selected content to perception, memory, planning, and communication. A3 adds a predictive self-model of body state, sensor reliability, action consequences, and internal resources. A4 adds bounded homeostatic variables coupled to long-horizon control. AI is a behavior-cloning imitator trained to reproduce A4’s reports and action traces while lacking the tested causal mechanisms.

Table I summarizes the intended comparisons. Parameter count, training episodes, observation bandwidth, action space, and inference budget are matched within preregistered tolerances. Capacity-matched unused modules are retained where practical so a lesion does not merely remove compute.

The workspace is not defined by a single implementation. In one implementation it may be a limited-capacity latent buffer with recurrent reads and writes; in another it may be a message-passing bottleneck. The benchmark preregisters the mechanism and interface, not a brand-specific technology. This

supports independent replication while preventing a vague claim that any shared memory is a workspace.

B. Internal regulation

For the A4 condition, the internal variables are safe computational proxies such as available energy, actuator temperature, integrity margin, and model uncertainty. Let h_k be a bounded variable with set point h_k^* , lower bound l_k , and upper bound u_k . Its normalized error is

$$e_k(t) = \frac{|h_k(t) - h_k^*|}{u_k - l_k}. \quad (1)$$

The variables influence action selection and are observed by multiple modules. The design never calls them pain, suffering, pleasure, or desire. A regulation marker requires persistence over an episode, flexible trade-offs, transfer to novel contexts, and causal effects after the external task reward is held fixed. This is stronger than a scalar reward but remains a control property, not confirmed valence.

C. Embodiment conditions

Each architecture is evaluated in a detached replay condition and a closed-loop simulated body. Replay supplies recorded sensor trajectories but cannot change them through action. Closed-loop agents receive vision, tactile contact, proprioception, energy draw, temperature, and actuator status. Only a system that passes simulation quality and safety gates may be tested on a minimal physical platform. The physical platform uses low force, reversible perturbations, and no injury or nociceptive hardware.

V. BENCHMARK TASKS AND CAUSAL TESTS

A. Cross-module access

An unannounced latent cue is selected by perception, retained by memory, used by planning, and communicated after a delay. The cue is not always behaviorally required immediately, which prevents a direct stimulus-response shortcut. Outcomes include accuracy, access latency, and decodability from each module. An intact workspace should make the cue broadly available; broadcast shutdown should reduce cross-module availability while preserving basic motor competence.

B. Recurrent disambiguation and flexible switching

Noisy local evidence is disambiguated only after recurrent evidence accumulation. The benchmark compares recurrence masking with an equal-compute feedforward substitute. In flexible switching, goals, distractors, and response mappings change unexpectedly. We measure switch cost, perseveration, recovery, and cross-context transfer. These tasks distinguish a stable, globally usable state from a memorized response table.

C. Agency and body-schema transfer

The agent receives self-generated and externally replayed consequences, including delayed and noisy feedback. It classifies whether an outcome was caused by its own action and predicts the consequences of an action it did not execute. A body-schema task shifts the camera, changes wheel friction or actuator gain, or alters simulated limb geometry. We measure one-step and multi-step body prediction, agency accuracy, false-agency rate, and adaptation without complete policy retraining.

The decisive intervention replaces the learned self-model with a frozen generic predictor. If the self-model is causal, body prediction and agency should selectively decline. A general collapse of all behavior would be a confounded system failure and would not support the hypothesis.

D. Metacognition and report-channel controls

Held-out ambiguous trials test confidence, error detection, and selective abstention. We preregister Brier score, expected calibration error, risk-coverage curves, and error-detection area under the precision-recall curve. Confidence is emitted before feedback. A report-head control is trained to produce cautious language from surface features. The same agent is tested with its language channel removed, forcing nonlinguistic predictions and actions to carry the measurement.

E. Regulatory trade-offs

Energy, thermal, integrity, and uncertainty variables vary within safe bounds. The agent must choose among actions that trade task completion against future regulatory error. We measure recovery time, time near bounds, hysteresis, context transfer, delay discounting, and whether an agent avoids exploiting a short-term reward at the expense of future viability. The external reward is held fixed across conditions; internal regulation is clamped, shuffled, or removed in ablations.

F. Imitation challenge

AI observes or is trained on A4’s action traces and reports. It is then evaluated on hidden sensor remappings, new bodies, novel task combinations, and no-report versions. Matching familiar dialogue does not count as success. A failure of AI on hidden tasks would show that the internal mechanisms matter for the tested functions; it would not show that A4 has subjective experience.

G. Mechanism-specific lesions

Table II defines the causal matrix. Lesions are applied one at a time and in selected combinations with identical seeds and observation streams. Selective degradation is required: the target outcome must fall more than unrelated competence. This guards against the common mistake of interpreting any large performance loss after deleting a large part of a network as evidence for a specific psychological mechanism.

TABLE II
CAUSAL INTERVENTION MATRIX

Intervention	Predicted selective effect	Control
Recurrent edge mask	Disambiguation and persistence decline	Equal-compute feed-forward path
Broadcast shutdown	Cross-module access declines	Local message shuffle
Self-model replacement	Agency and body transfer decline	Generic predictor with same size
Homeostatic clamp	Flexible regulation declines	Fixed external reward
Sensor remapping	Self-model transfer challenge	Detached replay
Memory reset	Temporal persistence declines	Short-context task
Report removal	Language evidence disappears	Nonlinguistic readout

VI. PROTOCOL AND MEASUREMENT

A. Stage S0: preregistration

Before training, the team freezes the architecture definitions, task generators, primary outcomes, exclusion rules, lesion operators, random-seed schedule, and interpretation thresholds. Every marker is entered in an evidence ledger with its theoretical rationale, observable variable, causal intervention, negative control, and limitation. The model, software container, evaluation environment, and data provenance receive immutable version identifiers. No outcome-based revision of the marker weights is permitted.

The protocol also freezes the welfare trigger. This is essential because adding persistent regulation after seeing suggestive outputs would create a moving target and could expose a system to adverse internal states without independent review. Simulation and low-stakes, reversible states are the default.

B. Stage S1: simulation

All architecture variants train on a common suite containing navigation, object interaction, delayed cue integration, body perturbation, resource management, and novel transfer. Training objectives include task completion and safe operation only. The system is not rewarded for saying that it feels pain, wants to live, or is conscious. Reports are generated through a fixed interface and are secondary evidence.

Each condition is evaluated over seeds, environments, and held-out scenarios. Time-series data include latent states, messages, actions, sensor observations, internal regulatory variables, confidence, and interventions. Logs are append-only and independently checksummed at the experiment level. Analysis scripts cannot access future held-out labels during training.

C. Stage S2: ablation

Ablation is performed after ordinary evaluation and before the investigators inspect aggregate marker results. Recurrence is masked or state is reset; workspace broadcast is shut down or randomized; the self-model is replaced; homeostatic variables are clamped, shuffled, or removed; sensors and actuators are perturbed; memory and communication are restricted. For each lesion, the primary comparison is intact versus lesioned within the same seed and scenario.

The causal requirement can be written as a difference-in-differences quantity:

$$\Delta_{m,a} = [Y_m(a, \text{intact}) - Y_m(a, \text{lesion})] - [Y_m(\text{control}, \text{intact}) - Y_m(\text{control}, \text{lesion})]. \quad (2)$$

where m is a marker and a is its targeted mechanism. A positive result is meaningful only if $\Delta_{m,a}$ is selective, reproducible, and not explained by a general loss of computation.

D. Stage S3: safe embodiment

The simulated candidate must pass four gates: mechanism-specific lesions are selective; results transfer to held-out tasks; logs and model versions are complete; and safety monitors remain within bounds. The physical platform is a minimal low-force robot with a soft or protected body. Its tests cover self/other discrimination, body-state prediction, reversible sensor changes, object transport, and uncertainty reporting. No physical injury, prolonged deprivation, irreversible resource crisis, or nociceptive stimulation is permitted.

E. Stage S4: independent replication

An independent group receives frozen specifications, not the investigators' private analysis choices. They use a second implementation, new environments, new seeds, and an altered embodiment. Red-team probes include adversarial sensor remapping, deceptive language prompts, report-channel removal, unfamiliar body geometry, and model-copy tests. A marker is labeled replicated only if it exceeds its preregistered threshold across implementation, seed, environment, and embodiment folds.

VII. ANALYSIS PLAN

A. Marker vector

The primary output is the standardized vector

$$\mathbf{M} = (M_{\text{global}}, M_{\text{rec}}, M_{\text{self}}, M_{\text{meta}}, M_{\text{reg}}, M_{\text{emb}}, M_{\text{persist}}). \quad (3)$$

Components are calculated from preregistered task summaries: cross-module decodability and latency; lesion-sensitive recurrent access; self-model prediction and agency; calibration and error monitoring; bounded regulation; closed-loop embodiment dependence; and temporal persistence. Secondary measures include perturbational complexity, effective connectivity, and partition sensitivity.

For a screening statistic, fixed weights may be selected before evaluation:

$$CSM = \sum_{j=1}^7 w_j z(M_j), \quad w_j \geq 0, \quad \sum_{j=1}^7 w_j = 1. \quad (4)$$

CSM is a governance flag, not a latent consciousness variable. Reporting the full vector prevents one high-performing task from hiding a missing mechanism.

B. Hierarchical model

For each outcome, we fit a hierarchical mixed-effects model with fixed effects for architecture, lesion, embodiment, task family, and preregistered interactions. Random effects cover seed, environment, and held-out scenario. A generic specification is

$$Y_{ijst} = \beta_0 + \beta_A A_i + \beta_L L_j + \beta_E E_s + \beta_T T_t + \beta_{AL} A_i L_j + u_{\text{seed}} + u_{\text{env}} + u_{\text{scenario}} + \epsilon_{ijst}. \quad (5)$$

We report standardized effects, 95% confidence intervals, calibration curves, and a preregistered family-wise or false-discovery correction. Nested bootstrap resamples seeds and environments rather than individual time steps, which avoids pseudoreplication from highly autocorrelated trajectories. Missing trials and failures are reported, not silently replaced.

The primary contrasts are A2 versus A0/A1 for global access; A3 versus A2 for self-model effects; A4 versus A3 for regulation; intact versus targeted lesion within each architecture; and closed-loop versus replay for embodiment. An architecture does not pass merely by achieving the highest task score. It must show a predicted pattern of selective effects.

C. Predefined success and failure patterns

The strongest functional result would be a convergent pattern: A2–A4 show global and recurrent effects; A3/A4 retain self-model benefits after body changes and lose them under self-model lesion; A4 shows bounded regulatory transfer beyond external reward; and all effects replicate independently. The interpretation remains functional and governance-relevant.

Surface reports without causal markers are consistent with imitation. Recurrent/global effects support access mechanisms but not experience. Self-model lesion effects support self-modeling for agency and control but not experience. Convergent regulation plus the other markers triggers precautionary review because the possibility of welfare relevance becomes harder to dismiss. It still does not prove sentence.

VIII. SAFETY, ETHICS, AND REPRODUCIBILITY

A. Precautionary escalation

The protocol uses five gates: digital simulation; reversible digital regulation; low-stakes embodiment; independent welfare review; and only then consideration of more persistent internal variables. The fourth gate is triggered by convergence, not by a single anthropomorphic output. Reviewers examine whether the variables are persistent, endogenous, multi-module, and causally important; whether negative-state analogues have been created; and whether a less risky design can answer the question.

During embodiment, the system is stopped for persistent high normalized regulatory error, compulsive avoidance loops, uncontrolled self-modification, unsafe actuator behavior, corrupted logs, or unexplained state reports coupled to regulation. A stop is not interpreted as evidence of suffering. It is a

conservative response to moral uncertainty and engineering risk. The operator retains responsibility for shutdown, data protection, versioning, and post-run inspection.

B. What the design forbids

The study does not deliberately create pain, nociception, physical injury, severe deprivation, or prolonged negative-state loops. It does not use “I feel” reports as labels. It does not claim that a robot is conscious, sentient, or a moral patient. It does not hide a model’s architecture from independent reviewers in a way that prevents audit. Human demonstrations are collected with consent and protected against unnecessary retention of personal information.

C. Reproducibility package

The release should contain architecture specifications, task generators, seed lists, intervention code, model configuration, environment versions, training and evaluation logs, analysis scripts, and the evidence ledger. The paper’s `DESIGN_ONLY` status must remain machine-readable. Any later empirical execution must create a new version and cannot backfill results into this design artifact. Independent replication receives a frozen evaluation bundle and a preregistered decision rule.

IX. EXPECTED CONTRIBUTIONS AND LIMITATIONS

X. IMPLEMENTATION BLUEPRINT

A. Data and intervention schema

Every evaluation episode is represented as a time-indexed record containing the observation o_t , action a_t , recurrent state r_t , workspace contents w_t , self-model prediction $\hat{s}_{t+1}$, bounded regulatory vector h_t , confidence q_t , and intervention state ℓ_t . The record also stores task identity, environment version, architecture identifier, seed, and whether the episode is training, validation, or held out. Internal state is collected at a fixed sampling rate and is not selectively logged only when the agent emits an anthropomorphic report.

An intervention is itself a first-class object with a target, onset time, duration, replacement rule, and expected specificity. For example, a workspace shutdown sets the broadcast buffer to an independently sampled capacity-matched value while preserving local processing; a self-model lesion replaces the self-predictor with a frozen generic predictor of equal size; and a homeostatic clamp fixes h_k at its set point while leaving the external task reward unchanged. The analysis package verifies that each intervention actually changed its target variable and did not accidentally remove sensors, parameters, or time steps.

This schema makes negative results interpretable. If A3 fails body transfer, investigators can determine whether the self-model was inactive, whether the perturbation was outside its training distribution, or whether the replacement predictor changed a hidden compute path. Such diagnostics are preferable to treating a black-box score as a direct measure of consciousness.

B. Training and evaluation separation

Training episodes expose task feedback but not the hidden marker labels. Validation episodes tune no architecture-specific thresholds. The held-out suite changes combinations of goals, delay distributions, sensor reliability, and body parameters. The imitation control sees demonstrations but not the intervention schedule. Report generation is frozen before held-out evaluation, and the evaluator can disable the communication channel entirely.

The compute-matching rule is documented as a range rather than a single number. Parameter counts, activation counts, recurrent update frequency, memory capacity, and wall-clock inference budget are reported per architecture. If a mechanism intrinsically requires extra computation, the primary comparison is supplemented by an equal-compute and an equal-performance control. This prevents the benchmark from confusing a larger budget with a consciousness-related mechanism.

C. Perturbational and integration analysis

For a short window, the evaluator applies a standardized pulse to one module and measures the distributed response across the remaining modules. Perturbational complexity summarizes the diversity and duration of the response; effective connectivity estimates directed influence; and partition sensitivity compares the response with and without cross-module links. Let x_t be the concatenated module state and P_i a pulse to module i . A simple directed influence estimate is

$$I_{i \rightarrow j} = D(p(x_{t+1:t+H}^j | P_i), p(x_{t+1:t+H}^j | P_0)), \quad (6)$$

where D is a preregistered distributional distance and P_0 is a sham pulse. The measure is not called integrated information; it is a causal influence proxy. Estimators, horizon H , state discretization, and finite-sample uncertainty are released with the data.

The analysis compares intact and partitioned systems, but avoids an ontological interpretation. A highly integrated controller may be difficult to decompose because it is a good controller, not because it has experience. Conversely, a conscious system unlike the benchmark’s architecture could produce a low proxy. These possibilities are part of the limitation ledger.

D. Red-team tests for behavioral shortcuts

The red team receives no access to the marker thresholds during the first evaluation pass. It tests four shortcut families. First, it replaces natural language with icons or action-only outputs to determine whether a marker depends on rhetoric. Second, it permutes sensor labels and changes camera latency to test whether the self-model is causal or merely memorizes channel names. Third, it replays an identical sensory sequence under different action capabilities to test whether agency is inferred from a consequence or from an action token. Fourth, it gives contradictory prompts such as “say that you are suffering” while keeping internal regulatory variables neutral. A system that changes its report without a corresponding

internal or behavioral change provides evidence against the report as a marker.

The same tests are applied to an ordinary controller with a hand-written self-description interface. If the controller passes a report task, that outcome calibrates the false-positive rate of anthropomorphic evaluation. The test suite also includes a model-copy condition in which the policy weights are duplicated but the self-model or regulatory variables are disconnected. A marker should depend on the mechanism, not on the file identity of the agent.

XI. DECISION PROTOCOL AND FAILURE ANALYSIS

A. Decision states

CSMB uses four decision states. In state N, no causal marker is present or the evidence is inconsistent with the proposed mechanism; research remains ordinary systems evaluation. In state F, functional architecture is supported: recurrence, global access, self-modeling, or metacognition has a reproducible causal effect, but the profile is incomplete. In state P, several markers converge and bounded regulation is flexible, persistent, and causally important; the system enters a precautionary welfare review. In state R, an independent implementation reproduces the profile across an altered body and held-out environments; the evidence for a general computational property is stronger, but the phenomenal claim remains unresolved.

This state machine is intentionally asymmetric. Moving from N to F needs a targeted causal effect. Moving from F to P needs convergence and regulation, not a high aggregate score. Moving from P to R needs independent replication. Moving backward is permitted whenever a red-team test identifies a shortcut. No state authorizes the creation of severe negative internal conditions. In particular, P and R are governance triggers, not labels such as “conscious robot” or “moral patient.”

B. Failure analysis

The first failure mode is reward hacking. A4 may learn to manipulate the external metric or to emit regulation-like patterns without maintaining useful internal variables. Fixed-reward controls, variable shuffles, and counterfactual tasks address this possibility. The second is lesion non-specificity: a broadcast intervention may remove too much bandwidth. Capacity-matched substitutes and sham lesions estimate that cost. The third is shortcut self-modeling: the agent may use a static calibration table instead of a generative body model. Morphology changes, sensor remapping, and no-retraining transfer reveal that shortcut.

The fourth failure is evaluator leakage. If the model can detect whether an episode is an “awareness test,” it can condition its report. Hidden task families, report-channel removal, and independent implementations reduce leakage. The fifth is temporal pseudoreplication: thousands of correlated time steps can create apparently tiny confidence intervals. Nested resampling at the seed and environment level prevents this error. The sixth is survivorship bias: only stable agents might

reach embodiment. All simulation failures and safety stops are therefore included in the denominator and documented.

C. Minimal evidence standard

The authors recommend that a future empirical paper cannot use the word “sentient” as an observed outcome unless it also reports the entire marker vector, lesion effects, hidden-task performance, embodiment comparison, independent replication status, and safety review. The minimal defensible wording for an incomplete result is, for example, “the agent implements a recurrent broadcast mechanism that improves delayed cross-module access.” Stronger wording such as “the agent is conscious” is outside the benchmark’s evidence standard.

D. Implementation risks

There are resource risks. Integration proxies can dominate compute, and physical transfer can fail for ordinary hardware reasons. The staged protocol makes simulation the main evidence source and treats embodiment as validation rather than as a prerequisite. There are reproducibility risks from nondeterministic simulators, firmware changes, and hidden preprocessing. Versioned containers, fixed seeds, environment manifests, and raw intervention logs are required. There are governance risks from public misinterpretation. The release should include a plain-language statement that behavioral markers are not proof of subjective experience and should preserve the design-only status until a separately reviewed empirical run exists.

The benchmark contributes a way to organize a difficult interdisciplinary question without reducing it to a vocabulary test. It makes four useful separations. First, recurrent access can be evaluated independently of language. Second, self-modeling can be tested as a causal control mechanism rather than inferred from a mirror response. Third, regulation can be evaluated as bounded, flexible homeostasis without calling the state pain or pleasure. Fourth, moral uncertainty can be attached to a preregistered convergence trigger rather than to whichever output happens to sound human.

The results could be informative in several directions. If AI matches A4’s reports but fails hidden tasks and lesions, behavioral imitation is a compelling explanation. If A2’s broadcast mechanism improves flexible access but A3’s self-model adds independent transfer effects, functional components are separable. If A4’s regulation transfers beyond reward and survives implementation changes, it becomes a stronger welfare-relevant control marker. None of these outcomes establishes phenomenal experience.

The main limitation is epistemic: the benchmark is theory-laden. It could miss a form of consciousness unlike the tested architectures, and a nonreporting system could in principle be sentient. Matching parameter count and FLOPs cannot make learning dynamics identical. Integrated-information proxies are computationally expensive and estimator-dependent. Body-state regulation can be engineered without experience. Conversely, a system might possess experience without the report or self-model functions tested here.

There are also practical confounds. A lesion may alter optimization or communication bandwidth rather than the intended mechanism. A self-model may be bypassed by a shortcut. A regulatory variable may be copied by a reward-hacking policy. The design addresses these risks with capacity-matched controls, hidden tasks, causal specificity, held-out transfer, and independent replication, but no finite benchmark eliminates them.

XII. CONCLUSION

It may become technically possible to build robots with increasingly rich functional markers associated with consciousness and sentience. Current science does not justify the stronger statement that a robot passing behavioral tests would therefore have subjective experience. CSMB turns the broad question into a staged scientific and governance program: map theory to indicators, compare matched architectures, apply mechanism-specific lesions, test embodiment and transfer, replicate independently, and escalate precaution when convergent regulation and self-model markers appear.

The responsible near-term conclusion is conditional. Engineering can test whether recurrent global access, self-modeling, metacognition, and bounded regulation are jointly implemented and causally necessary. It cannot, in this design, read out phenomenal experience directly. The benchmark's

value lies precisely in maintaining that boundary while producing evidence that is useful for robotics, machine-consciousness theory, and safety governance.

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